

HAOS-IIP Ferron Extension: Testable Signatures

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Scope

This note converts Materials Bridge Line A into forward-facing, falsifiable prediction records. It is not a new theory, not a proof of HAOS-IIP, and not a claim that ferrons embody HAOS-IIP.

Calibration Source

Release 64.1 provides the bounded calibration point: real Figshare ferron / NbOI₂ data loads; 15/15 target-window spectral records are found near 3.13 THz; mean peak frequency is 3.13893 THz; mean spectral recoverability is 0.928455; published Fig. 2d target-band traces contain 8 traces and 216 time points; the published group-velocity proxies are 58.8 km/s at 96 nm and 100 km/s at 224 nm; the computed-from-time-trace STFT diagnostic gives mean envelope correlation 0.995318; and the unified persistence proxy is 0.957192.

The raw-grid boundary remains active: `raw_stft_time_frequency_grid_status=NO_STFT_DATA_FOUND`. Published target-band traces and computed STFT diagnostics are not raw time-frequency maps.

Field-Facing Predictions

Prediction	Testable readout	Falsifier
Polar-mode address selection	Layered polar ferroelectrics or related polar oxyhalides should show a narrowband coherent readout in a pre-registered target window around an independently measured polar mode.	If matched polar materials show only broadband emission, or if non-polar controls show the same target-window recoverability, the prediction fails for that material class.
Thickness / boundary propagation dependence	Target-band packet arrival time should increase with pump-probe separation, and group velocity should vary with thickness or confinement under matched excitation.	If controlled thickness series show no distance-ordered peak delay and no thickness dependence beyond uncertainty, the prediction fails for that material class.
Re-addressing after secondary perturbation	After a secondary pulse, mild defect scattering, or boundary perturbation below damage threshold, the readout should preferentially return to the original target frequency window.	If secondary perturbation repeatedly produces broadband thermalization or a new dominant frequency with no preferential recovery of the original target window, the prediction fails for that perturbation regime.
Cross-dataset narrowband control discrimination	Pre-registered target-window recoverability should separate coherent directional propagation records from broadband or non-propagating controls better than amplitude-only ranking.	If target-window recoverability does not distinguish coherent propagation records from controls better than amplitude-only baselines, the prediction fails for the selected dataset class.

Provisional Thresholds

Suggested first-pass thresholds are intentionally conservative and revisable: target peak in at least 80% of controlled records, linewidth proxy below 0.1 THz where estimable, mean spectral recoverability at or above 0.85, frequency stability at or above 0.95, and target-band recoverability remaining above 0.70 before visible propagation failure.

Boundary

These are field-facing diagnostic candidates. They do not claim new physics, a materials ontology, a universal harmonic grid, or proof of HAOS-IIP. The numbers above are provisional thresholds seeded by the ferron audit, not universal constants.